

Theory-Based Interpretability in Deep Knowledge Tracing via Grounded Transformers

Concepcion Labra ¹ , Olga C. Santos ^{2,*} 

¹ Department of Artificial Intelligence, Computer Science School, UNED, 28040 Madrid, Spain; clabra@dia.uned.es

² PhyUM Research Center, Department of Artificial Intelligence, Computer Science School, UNED, 28040 Madrid, Spain; ocsantos@dia.uned.es

* Author to whom correspondence should be addressed.

Abstract

Knowledge Tracing, which enables the estimation of how students' knowledge evolves as they interact with educational content, is a cornerstone of intelligent tutoring systems. Traditional approaches coexist with deep knowledge tracing models that deliver better predictions but suffer from a critical drawback: lack of interpretability, which is particularly problematic in educational contexts. To overcome this limitation, we propose gTransformer, a novel model that unifies the high predictive performance of deep learning with the intrinsic interpretability of traditional approaches like Bayesian Knowledge Tracing. Our design employs an encoder-decoder Transformer that enriches input sequences with parameter estimates from the interpretable model. Attention mechanisms integrate these embeddings into a latent context vector, which is projected into updated parameter values constrained to remain pedagogically grounded to educational constructs while incorporating rich temporal dependencies learned by the model. These enriched parameters then drive predictions through interpretable Bayesian logic. Experiments demonstrating state-of-the-art accuracy show that our model achieves superior diagnostic granularity by identifying student-specific parameters—such as initial knowledge and learning rates—that capture individual longitudinal contexts. By anchoring deep representations to concepts defined by an intrinsically interpretable reference model, gTransformer offers a pedagogically interpretable alternative for data-driven personalization.

Keywords: deep knowledge tracing; theory-based interpretability; grounded transformers; bayesian knowledge tracing; educational data analysis; personalized learning

1. Introduction

Knowledge Tracing (KT) constitutes a foundational problem in Artificial Intelligence in Education and Intelligent Tutoring Systems. It defines the task of modeling the dynamic evolution of a student's knowledge state over time based on their history of interactions with learning materials [1]. By estimating the probability that a student has mastered a specific skill at a given moment, KT systems enable the delivery of personalized instruction—recommending remedial content for struggling students or accelerating the curriculum for those who demonstrate proficiency. In an era where educational environments are becoming increasingly digital and diverse, the ability to accurately track and interpret student mastery is not merely a technical optimization but a critical requirement for effective, scalable, and learner-centred education.

Received:

Revised:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *Journal Not Specified* for possible open access publication

under the terms and conditions of the

[Creative Commons Attribution](#)

(CC BY) license.

Historically, the development of KT has been characterized by a dichotomy between two distinct paradigms [2]. The first, which predominated in the field for years, is characterized by probabilistic models such as Bayesian Knowledge Tracing [1] and Factor Analysis Models based on logistic regression—including Additive Factor Models [3], Performance Factor Analysis [4], and Knowledge Tracing Machines [5]. Factor Analysis Models are theoretically supported by the Item Response Theory (IRT) [6], which has played a large role in educational assessment and measurement. These traditional approaches are intrinsically interpretable: they model learning as a process governed by explicit parameters related to pedagogical, cognitive or psychometric principles. Because the model structure mirrors a theoretical understanding of human learning, educators can inspect these values to validate the model’s reasoning and trust its outputs. However, this transparency comes at the cost of representational rigidity; the simplifying assumptions required by these models often limit their predictive power, causing them to struggle with the complex, non-linear, and long-range dependencies inherent in real-world student behavior [7].

The second paradigm emerged with the advent of Deep Knowledge Tracing (DKT) [8], which leverages deep learning techniques ranging from initial Recurrent Neural Networks to more recent Transformer architectures [9]. The latter, based on encoder-decoder components and attention mechanisms, are especially relevant as they provide the foundation for current Large Language Models. By treating student interaction histories as temporal sequences, Transformer models apply techniques similar to those that characterize LLMs. DKT models have achieved state-of-the-art predictive performance, significantly outperforming classical baselines due to a structural high capacity that allow them to extract latent features from vast amounts of data [2]. However, this substantial gain in predictive performance has come at the cost of model transparency, creating a significant interpretability gap. Deep learning models are notoriously opaque “black boxes”, where the internal high-dimensional representations bear no direct correspondence to constructs that are interpretable in terms of human-understandable concepts or established domain knowledge. In a high-stakes domain such as education—where algorithmic decisions influence learning trajectories, grades, and opportunities—this lack of transparency poses critical challenges for practical adoption.

Extensive research has sought to bridge this gap, with Bai et al. [10] classifying modalities into *post-hoc* and *ante-hoc* categories. While *post-hoc* methods aim to extract explanations from black-box models after training, *ante-hoc* approaches seek to achieve intrinsic interpretability by design, ensuring the model’s internal logic is transparent from the outset. Within this latter category, recent models have attempted to achieve intrinsic transparency by integrating attention mechanisms [11,12] or by incorporating side information such as Item Response Theory (IRT) parameters [13–15]. These efforts aim to make the decision-making process visible, moving away from pure black-box architectures toward systems whose internal states can be probed or visualized.

However, from the perspective of domain experts such as teachers and pedagogical designers, these solutions frequently fall short. Post-hoc explanations often rely on technical relevance scores or local approximations that remain disconnected from established pedagogical principles [16] while most current ante-hoc methods do not produce outputs that map directly to human-understandable educational constructs. As noted in various surveys [10,17], when deep learning models incorporate domain knowledge, the primary objective uses to be the optimization of predictive performance, while interpretability is often overlooked or left as an unaddressed concern.

Consequently, there is a critical need for new approaches that move beyond technical transparency toward true pedagogical interpretability. For a Knowledge Tracing system to be confidently adopted in high-stakes educational environments, its estimations must be

grounded in trusted and empirically validated principles derived from relevant fields such as pedagogy, cognitive science, or psychometrics. Educators require models that speak their language—tracking mastery not through opaque latent features, but through concepts explicitly defined by established frameworks, including the skills, prior knowledge, or learning rates postulated by traditional approaches such as BKT and Factor Analysis Models.

To address this challenge, we introduce the *grounded Transformer (gTransformer)*, a variant of attention-based Transformers that implements a novel *Latent Grounding* methodology to bridge the gap between deep learning expressiveness and theoretical alignment. This approach explicitly anchors the Transformer’s latent representations to the semantic constructs of an intrinsically interpretable reference model. The gTransformer architecture utilizes an encoder-decoder design with attention mechanisms to create context vectors that encode complex student interaction patterns. These vectors are projected into enriched parameters, which then drive predictions using the reference model’s interpretable logic. By doing so, the model’s estimations can be tracked down to domain knowledge and theoretical principles. This architecture satisfies the requirements for intrinsic interpretability by providing “grounded interpretability” by design, as the estimations are expressed through the same well-validated concepts used in established interpretable systems.

Our approach contributes to bridging the dichotomy between high-capacity deep learning and domain-specific theoretical transparency, providing two practical advantages. First, compared to non-interpretable DKT models, gTransformer offers intrinsic pedagogical transparency by generating actionable parameters—such as prior knowledge and learning rates—that are directly meaningful to educators. Second, compared to traditional approaches, gTransformer is context-aware, i.e it is able to encode individual patterns taking into account the history of student interactions, overcoming the limitations of rigid population-level parameters and Markovian assumptions characteristic of BKT. For instance, it can distinguish between a learner requiring foundational remediation and one exhibiting high acquisition momentum even when their immediate local performance appears indistinguishable. By integrating these capabilities, gTransformer provides a foundation for truly individualized student-centered instruction, allowing educators to move beyond predicting *if* a student will succeed toward understanding *why* they are struggling.

We evaluate our approach across four primary dimensions. First, we demonstrate parameter recovery accuracy by computing Pearson correlations and generating scatter plot visualizations between the model’s predicted parameters and theoretical targets from a fitted BKT model, achieving correlations exceeding 0.7 for both initial mastery and learning rate estimates. Second, we conduct ablation studies to quantify the interpretability cost, showing that theory-guided grounding achieves full interpretability with less than 0.5 percentage points of predictive accuracy loss. Third, we employ a dual evaluation protocol that separately measures both standard supervised predictions and interpretable predictions, demonstrating that the latter outperform classical BKT AUC by 10.8% while remaining highly competitive with non-interpretable DKT baselines. Finally, we present case study visualizations of learning trajectories across distinct learning situations characterized by different combinations of student prior knowledge and learning rates. These analyses show how gTransformer’s predictions are context-aware, effectively incorporating the history of interactions in a way that Markovian models like BKT are incapable of. By capturing these behavioral nuances, the model provides longitudinal, individualized diagnostics that support pedagogical decisions such as appropriate placement, pacing, and targeted intervention.

Our research is guided by three primary research questions:

- **RQ1: Theory-Based Interpretability Through Grounded Transformers**

Can deep knowledge tracing models achieve state-of-the-art predictive performance while providing interpretability grounded in established principles and theories?	133
Specifically, can we design a transformer architecture that produces pedagogically meaningful mastery estimations that are explainable through a causal and interpretable logic, such as Bayesian Knowledge Tracing?	134
• RQ2: Trade-Offs Between Predictive Performance and Interpretability	135
How do the metrics of the supervised, interpretable, and BKT predictions compare?	136
What is the cost of interpretability in terms of AUC? How much predictive gain do the interpretable grounded predictions achieve compared to traditional BKT?	137
• RQ3: Practical Value for Student-Centered Personalization	138
Beyond providing interpretable diagnostics, does the high capacity of gTransformer to capture intricate interaction patterns offer advantages over traditional models? Specifically, can these capabilities be leveraged to enhance student-centered personalization relative to population-based models such as Bayesian Knowledge Tracing?	139
The remainder of this paper is structured as follows. Section 2 reviews related work, while Section 3 describes the gTransformer model, detailing the construction of embeddings, the grounding mechanism, and the prediction logic. Section 4 presents the experimental setup, followed by a discussion of the empirical results in Section 5. Finally, Section 6 provides the concluding remarks.	140

2. Related Work

2.1. Bayesian Knowledge Tracing

For this study, we selected BKT [1] as our reference theoretical framework. It models the learning process as a hidden Markov process governed by four parameters providing a causal narrative of student performance that is intuitive to educators and aligned with cognitive science principles:

- *Initial Mastery*, the probability a student knows a skill prior to practice;
- *Learn Rate*, the probability of transitioning from unlearned to learned states after a practice opportunity;
- *Guess*, the probability of a correct response despite a lack of mastery; and
- *Slip*, the probability of an incorrect response despite mastery.

Standard BKT suffers from a significant limitation: it typically operates at a population level, estimating a single set of parameters for all students interacting with a given skill. While *Individualized BKT* approaches exist, they are often computationally expensive and struggle with data sparsity [18]. The Representational Grounding used in the gTransformer model offers a new way to address individualization by leveraging the capabilities of Transformers to compress student interaction history into a context vector. This allows the model to infer personalized initial knowledge and learning rates dynamically, transforming population-level BKT insights into high-granularity student diagnostics.

2.2. Deep Knowledge Tracing

The application of deep learning to knowledge tracing was initiated by the Deep Knowledge Tracing (DKT) model [8], which framed student performance as a sequence modeling task. While pioneering, DKT was constrained by the inherent limitations of Recurrent Neural Networks (RNNs). These architectures are difficult to train [19] and their sequential nature precludes parallelization within training examples, a bottleneck that becomes critical at longer sequence lengths where memory constraints limit efficiency[9].

The introduction of attention-based architectures, such as Self-Attentive Knowledge Tracing (SAKT) [11] and Separated Self-Attentive Neural Knowledge Tracing (SAINT) [19]

[20], marked a significant shift toward the attention-based Transformers that characterizes modern Large Language Models. They leverage this architecture ability to capture long-range dependencies, allowing for a non-Markovian representation of a student's entire history. More recently, Attentive Knowledge Tracing (AKT) [12] further refined this by incorporating skill difficulty and context-aware attention mechanisms that weigh past interactions based on their temporal distance.

Beyond the prominent recurrent and attention-based paradigms, the research community has explored several other architectural directions to capture the multifaceted nature of student learning. Notable categories include Memory-Augmented models that utilize external storage to explicitly track concept mastery, Graph-Based approaches that model the topological relationships between skills, and specialized variations such as Text-Aware or Forgetting-Aware models that incorporate exercise semantics and temporal decay into the tracing process [2]. While these innovations have further enhanced predictive accuracy, they share the common challenge of balancing model complexity with the pedagogical transparency required for meaningful educational intervention.

2.3. Interpretability in Deep Knowledge Tracing

Interpretability in DKT remains an open challenge despite a variety of attempts, which can be categorized into two primary modalities: *post-hoc* and *ante-hoc*. Post-hoc methods attempt to explain trained black-box models by utilizing either model-specific techniques, such as Layer-Wise Relevance Propagation (LRP) [21], or model-agnostic approaches like Local Interpretable Model-agnostic Explanations (LIME) [22]. However, Rudin [16] argues that relying on post-hoc explanations for high-stakes decision-making is fundamentally flawed. For teachers and domain experts, these methods offer limited utility as they produce explanations tied to technical model structures rather than pedagogically sound principles, necessitating significant technical expertise for meaningful interpretation.

The second category, *ante-hoc* methods, pursues intrinsic interpretability through two primary approaches. The first consists of traditional models that rely on transparent structures, such as BKT or Factor Analysis Models. The second encompasses DKT variants that achieve interpretability by incorporating additional modules or utilizing side information [10]. Among these, attention mechanisms have emerged as the most prominent architectural extensions. They have been integrated into a wide range of models, from the pioneering Self-Attentive Knowledge Tracing (SAKT) [11] to more recent architectures like Attentive Knowledge Tracing (AKT) [12], which achieves high predictive performance by utilizing multi-scale attention to capture interaction patterns across varying temporal resolutions. However, relying solely on attention weights presents similar practical obstacles to those of post-hoc methods for educational practitioners, as these mechanisms do not provide an explicit interpretation grounded in pedagogically sound principles.

The second approach to ante-hoc interpretability involves the integration of side information, with Item Response Theory (IRT) [6] serving as the most prominent foundation. IRT estimates the probability of a correct response based on a student's latent ability and the item's difficulty. This theory has been coupled with DKT models such as Deep-IRT [13], which utilizes dynamic key-value memory networks to capture learners' trajectories and uses inferred abilities and difficulties to predict performance. Similarly, TC-MIRT [14] embeds multidimensional IRT parameters to predict student states and generate interpretable parameters. Other relevant models include KIKT [23], which harnesses IRT to simulate student performance; LANA [24], which adopts an IRT variant for ability clustering; and QIKT [15], which features question-centric representations integrated with an interpretable IRT layer. Beyond IRT, researchers have leveraged diverse principles, such

as constructive learning [25,26], learning and forgetting curves [27], finite state automata (FSAs) [28], classical test theory (CTT) [29], and monotonicity constraints [30].

In contrast to these models, whose diagnostic utility is often restricted by the specificity and scope of the single underlying theory serving as side information [10], gTransformer offers a more flexible framework. It can be grounded in any established theoretical reference, effectively decoupling the expressiveness of deep learning from the constraints of specific pedagogical models.

2.4. Informed Machine Learning

The integration of domain knowledge into deep learning has been broadly explored through conceptual frameworks such as Theory-Guided Data Science (TGDS) [31], Physics-Informed Machine Learning (PIML) [32], and Informed Machine Learning (IML) [17].

TGDS emerged as a paradigm to address the limitations of purely data-driven “black-box” models, which may produce results inconsistent with established domain knowledge [31]. By explicitly requiring scientific consistency, TGDS integrates theoretical insights into the learning process to ensure that models remain scientifically plausible. This integration is typically achieved via theory-guided learning—where domain-specific invariants serve as regularization terms—or through the design of neural architectures that respect the structural constraints of the problem [31,33]. A prominent implementation of this paradigm is Physics-Informed Neural Networks (PINNs) [32,34], which embed differential equations directly into the loss function, forcing the model to satisfy theoretical constraints while learning from empirical data.

A widely adopted technique to enforce this consistency involves incorporating domain-specific constraints into the loss function as follows [31,35]:

$$\mathcal{L} = \mathcal{L}_{TRN}(Y_{true}, Y_{pred}) + \gamma \mathcal{L}_{THEORY}(Y_{pred}), \quad (1)$$

where $\mathcal{L}_{TRN}$ measures the supervised error between true labels Y_{true} and predicted labels Y_{pred} , while $\mathcal{L}_{THEORY}$ penalizes violations of theoretical constraints, weighted by the hyperparameter γ .

Von Rueden et al. [17] formalized these efforts into an IML taxonomy, which categorizes knowledge integration by its source, representation, and the stage of the machine learning pipeline where it is incorporated. Knowledge can be represented as algebraic equations, logical rules, or probabilistic priors, and integrated at various stages: within the training data, through the design of tailored architectures, or by influencing the model’s final output.

We operationalized these taxonomic dimensions in gTransformer by integrating theoretical priors from an interpretable BKT model into the input layer and employing a multi-objective loss function to anchor the model’s predictions to BKT parameters. However, while most IML approaches focus on enhancing predictive performance [17], gTransformer aims to achieve theory-based interpretability by grounding estimations in sound pedagogical concepts, as detailed in Section 3.1.

3. The gTransformer Model

3.1. Theory-Guided Information Flow

Figure 1 illustrates the information flow through the system, mapped to the taxonomic dimensions of IML as defined in [17].

Within the context of our architecture, these dimensions are operationalized as follows:

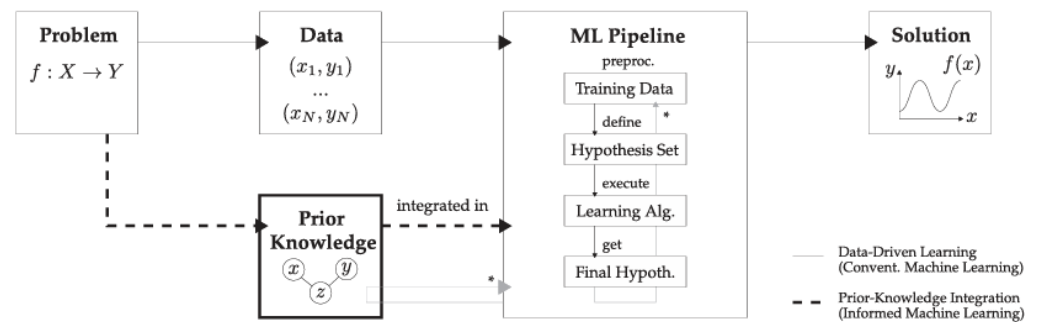


Figure 1. Information flow in gTransformer, following the taxonomy proposed by Von Rueden et al. [17].

- **Prior Knowledge:** The source of theoretical information comprises simulation results from a reference model, typically expressed as parameter values that represent foundational pedagogical constructs.
- **Training Data:** Empirical student interaction histories are augmented with theoretical priors to guide the learning process toward pedagogically valid states.
- **Hypothesis Set:** An attention-based Transformer architecture is employed to provide the inductive biases necessary for grounding internal latent representations within the conceptual framework of the reference model.
- **Learning Algorithm:** The model is optimized via a multi-objective loss function formulated to balance the trade-off between predictive performance and pedagogical interpretability.
- **Final Hypothesis:** A set of learned weights that drive latent representations from which parameter values aligned with the reference model can be extracted.

3.2. Architecture

The core architecture of gTransformer, illustrated in Figure 2, builds upon the standard Transformer framework [9] by integrating specialized modules designed to achieve grounded interpretability. It is structured into the following functional layers:

1. *Input Components:* the data flow begins with the ingestion of student interactions enriched with parameters from a *BKT Reference Model*. These per-skill parameters, prior knowledge $P(L_0)$ and learning rate $P(T)$, serve as the theoretical anchor to ground the latent representations p_{l0} and p_t . The interactions can be considered as sequences of tuples (x, y) , where x is the skill or concept id and y is a boolean value indicating if the student's response was correct or incorrect. Additionally we can add the question id q to obtain the sequence of (q, c, r) tuples showed in the Figure 2.
2. *Individualized Embeddings:* (x, y) are transformed into *individualized embedding vectors* (x', y') . As detailed in Section 3.3, these representations decompose generic ids into nuanced vectors that capture task-specific difficulty and student-level behavioral traits, providing the multi-dimensional material necessary for attention-driven context discovery.
3. *Contextual Transformer Architecture:* the core engine utilizes an *encoder-decoder Transformer* stack. The encoder processes the student's longitudinal interaction history into a tensor of latent representations. For each timestep t , the decoder then processes this history via cross-attention, resulting in a dense *context vector* ($z_t \in \mathbb{R}^d$) that encapsulates the student's behavior.
4. *Grounded Parameter Projection:* the dense context vector z_t is projected into dynamic contextual displacements through its interaction with learnable semantic axes. These

displacements are added to the theoretical base parameters $p_{L_0}^{\text{base}}, p_T^{\text{base}}$ provided by the reference model. This additive composition, yields the final individualized parameters p_{L_0}, p_T , ensuring that high-resolution behavior signatures remain structurally anchored to pedagogical theory.

5. *Dual Prediction Heads*: the system generates dual predictions through a *Supervised Output* that generates high-accuracy predictions p_{sup} from the Transformer context vector z_t . Simultaneously, the *Interpretable Output* uses the grounded parameters to drive a BKT logic block, producing the interpretable prediction p_{ref} which are grounded on the parameter values given by the BKT reference model.
6. *Multi-Objective Training Pipeline*: the architecture is optimized via a total loss $\mathcal{L}_{\text{total}}$ that balances predictive accuracy with theoretical grounding. This objective integrates a supervised loss $\mathcal{L}_{\text{sup}}$ and an interpretability loss $\mathcal{L}_{\text{ref}}$, both calculated as the Binary Cross Entropy (BCE) between the respective prediction heads p_{sup} and p_{ref} and the ground truth. Additionally, a linear probing loss $\mathcal{L}_{\text{probe}}$ —calculated as the Mean Squared Error (MSE) between diagnostic probes and theoretical targets—verifies the structural preservation of pedagogical constructs within the Transformer’s internal representations.

3.3. Embeddings

In standard educational datasets, such as ASSISTments 2009, ASSISTments 2015, Algebra 2005, and others [36], student interactions are recorded at the level of specific questions or tasks, each of which is associated with one or more underlying concepts or knowledge components. This structure reflects the fact that proficiency in a concept (e.g., the Pythagorean Theorem) is acquired through interactions with a diverse range of tasks. While all tasks involving a concept share the same semantic core, they differ in their specific manifestations—most notably in their intrinsic difficulty or complexity. Therefore, an effective representation must capture both the shared identity of the concept and the unique deviation of the specific task.

In Transformer-based models, we operationalize this principle by representing tasks as high-dimensional embedding vectors:

$$x' = c + u \cdot d \quad (\text{Task Embedding}) \quad (2)$$

In this formulation, c is the *Concept Base*, representing the invariant semantic identity of the knowledge component. The scalar u is the *Problem Difficulty* (u_q), and d is the *Difficulty Variation Axis* (d_c). This Rasch-inspired decomposition ensures that the embedding x' captures both the general concept and the specific task complexity.

Similarly, we represent student interactions as:

$$y' = e + u \cdot f \quad (\text{Interaction History Embedding}) \quad (3)$$

where e is the *Interaction Base* (encoding concept and response outcome), and f is the *Interaction Variation Axis*. These embeddings x' and y' serve as the primary queries and values for the Transformer’s attention mechanism.

Rather than using theoretical parameters as static inputs, gTransformer treats them as *grounded objectives* whose values are dynamically estimated from the Transformer’s context. To achieve this, we define a set of *Theoretical Bases* ($p_{L_0}^{\text{base}}, p_T^{\text{base}}$) and *Semantic Axes* ($\mathbf{k}_c, \mathbf{v}_c$) for each concept. The final individualized parameters are estimated at each timestep t as follows:

$$p_{L_0} = \sigma\left(p_{L_0}^{\text{base}} + (z_t \cdot \mathbf{k}_c) + k_s\right) \quad (4)$$

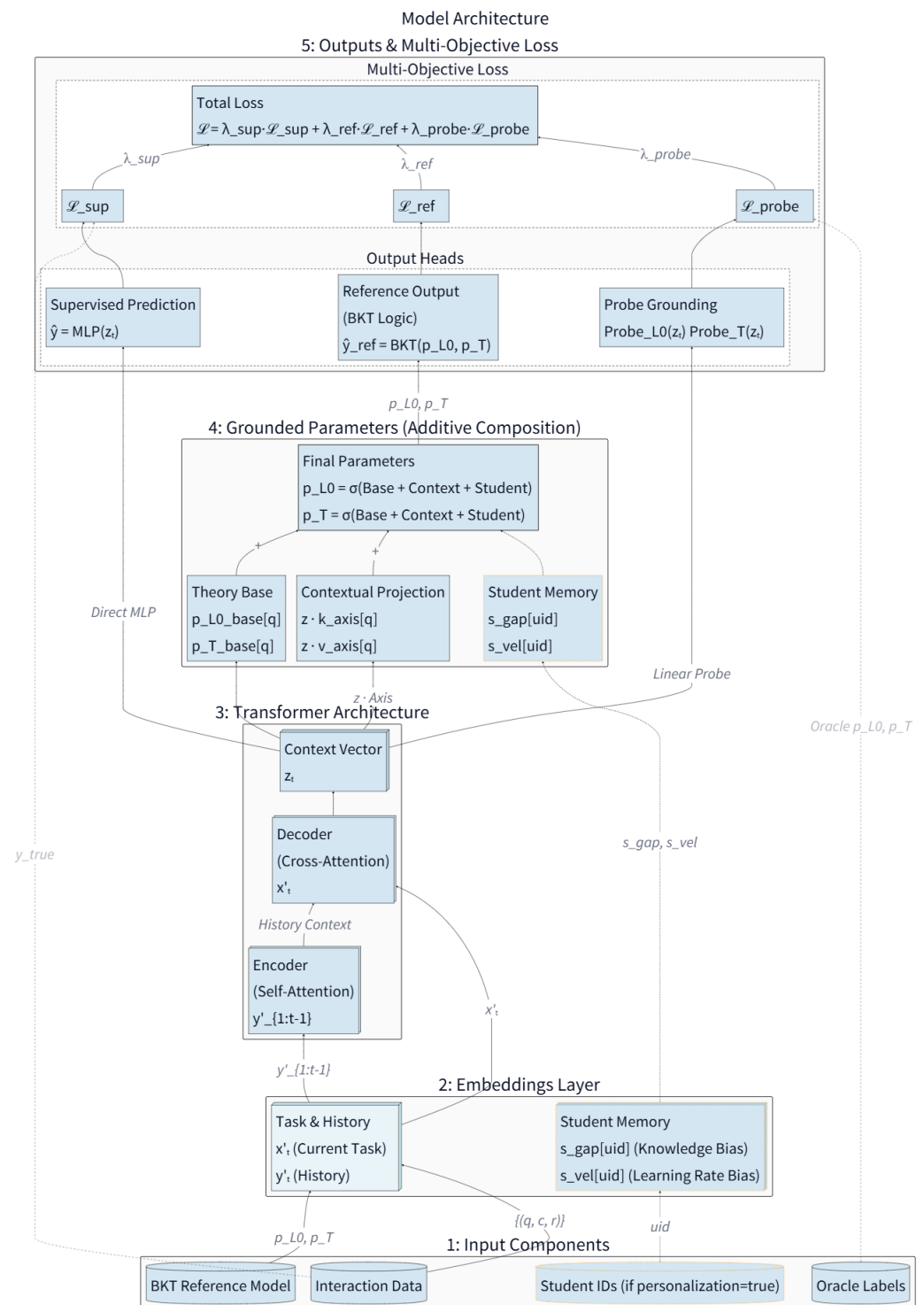


Figure 2. Functional layers of the gTransformer Architecture: (1) *Input Components*: student interaction data enriched with BKT reference parameters; (2) *Individualized Embeddings*: task-specific and student-specific latent transformations (x' , y'); (3) *Contextual Transformer Architecture*: encoder-decoder synthesis of longitudinal learning history into a context vector (z_t); (4) *Grounded Parameter Projection*: additive composition of theoretical bases and contextual displacements to derive pedagogical factors (p_{know} , p_{rate}); (5) *Dual Prediction Heads*: high-accuracy supervised forecasts (p_{sup}) and theoretically-consistent interpretable predictions (p_{ref}); and (6) *Multi-Objective Training Pipeline*: alignment of neural representations with pedagogical theory via supervised ($\mathcal{L}_{sup}$), interpretability ($\mathcal{L}_{ref}$), and probing ($\mathcal{L}_{probe}$) losses.

$$p_T = \sigma(p_T^{\text{base}} + (z_t \cdot \mathbf{v}_c) + v_s) \quad (5)$$

where $p_{L_0}^{\text{base}}$ and p_T^{base} are the theoretical priors initialized from the BKT reference model; $z_t \cdot \mathbf{k}_c$ and $z_t \cdot \mathbf{v}_c$ represent the *Contextual Projection*, where the latent state of the student is projected onto the semantic dimensions of knowledge mastery and acquisition rate; and k_s, v_s are the *Student Personalization Scalars*, capturing stable student-level traits that persist beyond specific interactions.

This design enables a pedagogical interpretation of the model's behavior:

- *Personalized Mastery* (p_{L_0}): Adjusts the theoretical prior by observing the student's behavior history, identifying cases where a student's prior knowledge significantly deviates from the population mean.
- *Contextual Learning Rate* (p_T): Modulates the acquisition speed based on the longitudinal signature of the student's interactions, distinguishing between fast learners and those requiring more intensive support even when their current performance is identical.

3.4. Loss Functions

The optimization of gTransformer is governed by a multi-objective loss function designed to ensure that the high-capacity Transformer architecture remains aligned with pedagogical principles. Following established patterns in theory-guided machine learning [31,35], we incorporate theoretical constraints as grounding terms in the optimization objective. This approach steers the model to learn latent representations that are both predictive of student outcomes and semantically consistent with established educational constructs. The total loss $\mathcal{L}_{\text{total}}$ is defined as a weighted sum of supervised, grounding, and regularization components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} + \lambda_{\text{probe}} \mathcal{L}_{\text{probe}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} \quad (6)$$

3.4.1. Supervised Alignment ($\mathcal{L}_{\text{sup}}$)

The primary objective $\mathcal{L}_{\text{sup}}$ uses standard Binary Cross-Entropy (BCE) between the gTransformer supervised predictions $p_{\text{sup},t}$ and the observed ground truth outcomes $r_t \in \{0, 1\}$. This loss ensures predictive accuracy by minimizing the deviance from observed student responses:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{N} \sum_{t=1}^N [r_t \log(p_{\text{sup},t}) + (1 - r_t) \log(1 - p_{\text{sup},t})] \quad (7)$$

3.4.2. Theoretical Grounding and Alignment

To satisfy the criteria for Pedagogical Interpretability, gTransformer anchors both its internal representations and its diagnostic behavior to BKT logic through two specialized grounding terms.

Interpretable Fidelity ($\mathcal{L}_{\text{ref}}$): To ensure the pedagogical parameters drive meaningful predictions, the model enforces behavioral alignment through the *Interpretable Output*. $\mathcal{L}_{\text{ref}}$ is calculated as the BCE between the BKT-logic predictions (p_{ref}) and the ground truth labels:

$$\mathcal{L}_{\text{ref}} = -\frac{1}{N} \sum_{t=1}^N [r_t \log(p_{\text{ref},t}) + (1 - r_t) \log(1 - p_{\text{ref},t})] \quad (8)$$

By penalizing the predictive deviance of the logic-constrained head, this grounding loss ensures that the inferred p_{L_0} and p_T factors retain their intended pedagogical utility.

Diagnostic Probing Loss ($\mathcal{L}_{probe}$): Finally, we implement an "Active Grounding" mechanism via linear probes. $\mathcal{L}_{probe}$ measures the MSE between dedicated probing heads (linear mappings from z_t) and the theoretical targets. This verifies that the Transformer's context vector explicitly encodes the targeted pedagogical signatures:

$$\mathcal{L}_{probe} = \sum_{k \in \{L_0, T\}} \text{MSE}(h_{probe,k}(z_t), p_k^{\text{base}}) \quad (9)$$

3.4.3. Regularization and Structural Diversity

While grounding losses anchor the global structure of the latent space, gTransformer ensures that student-level individualization remains *parsimonious* and concept representations remain *diverse*. This is achieved through a multi-term regularization objective $\mathcal{L}_{reg} = \lambda_{L2}\mathcal{L}_{L2} + \lambda_{div}\mathcal{L}_{div}$.

Parsimonious Individualization: we apply L_2 penalties to the scalar parameters governing student-level deviations and item difficulty:

$$\mathcal{L}_{L2} = \sum_{q \in Q} u_q^2 + \sum_{s \in S} (k_s^2 + v_s^2) \quad (10)$$

where u_q represents item difficulty variance, and $\{k_s, v_s\}$ are the student-specific shifts for proficiency and rate.

Semantic Diversity Loss: finally, to maintain the structural integrity of the semantic axes $(\mathbf{k}_c, \mathbf{v}_c)$, we incorporate a diversity loss $\mathcal{L}_{div}$. This term penalizes cosine similarity between the axes of different knowledge components, preventing the collapse of the latent space and ensuring that each concept maintains a unique pedagogical signature.

4. Experimental Setup

4.1. Implementation Details

We implemented the gTransformer model in PyTorch and used the benchmark library PYKT [36] to leverage standardized data preprocessing, dataset splitting and baseline models. For training, we used standard 5-fold cross-validation with an 80/20 train/test split. The model was trained using the Adam optimizer with a learning rate of $1e-4$, a batch size of 64, and a dropout rate of 0.2 to prevent overfitting. The maximum number of epochs was set to 200, with an early stopping mechanism (patience=10) to terminate training if validation performance plateaued.

The Transformer architecture was configured with an embedding dimension d_{model} of 256, 8 attention heads, and 4 encoder/decoder blocks. The feed-forward dimension d_{ff} was set to 512. Regularization penalties for individualization parameters (L_2 on u_q, k_s, v_s) were all set to $1e-5$. For reproducibility, all experiments were seeded (seed=42) and executed on NVIDIA A100 GPU infrastructure. Predictive performance was evaluated using Area Under the ROC Curve (AUC) and Accuracy (ACC), while interpretability was assessed using the metrics defined in Section 5.2.

4.2. Datasets

We evaluated gTransformer across four widely used educational datasets following the S-protocol for sequence length consistency [36]:

- ASSISTments 2009 (AS2009): A benchmark dataset from the ASSISTments platform (2009-2010), representing a standard for knowledge tracing evaluation [37].
- ASSISTments 2015 (AS2015): A large-scale release from the same platform with a high volume of student records [37].

- Bridge to Algebra 2006 (Bridge2006): A dataset from the KDD Cup 2010 EDM Challenge focused on the Carnegie Learning Bridge to Algebra system [38].
- NeurIPS 2020 Education Challenge (NIPS34): A diagnostic dataset from the Eedi platform containing multiple-choice math questions [39].

5. Results and Discussion

5.1. Predictive Performance

To evaluate the predictive performance of the gTransformer model, we used the Area Under the Curve (AUC) and the Classification Accuracy (ACC) of the models on the test set with the 5 datasets described in Section 4.2. The results are shown in the Table 1.

Table 1. Predictive Performance across the evaluated datasets (Mean AUC and Mean ACC).

Model	AS2009		AS2015		Bridge2006		NIPS34	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
gTransformer	0.8240	0.7573	0.7224	0.7520	0.7879	0.7565	0.7501	0.6868
AKT	0.8271	0.7620	0.7246	0.7540	0.7934	0.6947	0.7745	0.7081
ATKT	0.8286	0.7579	0.7755	0.7637	0.6911	0.6395	0.7391	0.6783
DKT	0.8208	0.7547	0.7268	0.7536	0.6701	0.6209	0.7354	0.6772
DKT+	0.8216	0.7570	0.7284	0.7544	0.6693	0.6121	0.7362	0.6781
DKVMN	0.8082	0.7476	0.7174	0.7535	0.6741	0.6522	0.7208	0.6646
SAINT	0.7638	0.7207	0.7031	0.7483	0.7340	0.7152	0.7127	0.6563
SAKT	0.6955	0.6608	0.6810	0.7405	0.5627	0.6328	0.6686	0.6234

When comparing these results with baseline architectures, we observe that gTransformer achieves highly competitive performance across all benchmarks. In terms of Mean AUC, gTransformer consistently ranks among the top performers, securing the 2nd position on the Bridge2006 and NIPS34 datasets—surpassed only by AKT model—and the 3rd position on AS2009. Notably, in terms of Classification Accuracy (ACC), gTransformer outranks all baseline models on the Bridge2006 dataset, achieving the highest value of 0.7565. These results provide empirical evidence that the models’ pedagogical grounding does not compromise its predictive integrity, maintaining state-of-the-art performance while providing additional layers of interpretability. Furthermore, to test the hypothesis that interpretability constraints might degrade accuracy, we evaluated an *unconstrained* variant of gTransformer (with all grounding loss terms λ_{ref} , λ_{init} , λ_{rate} set to zero). On the primary AS2009 dataset, this unconstrained baseline achieved a Mean AUC of 0.8065, significantly lower than the grounded model’s 0.8240. This result suggests that the pedagogical regularization acts as a beneficial inductive bias, guiding the optimization toward more generalizable minima rather than hindering performance.

5.2. Interpretability Validation (RQ1)

To verify that gTransformer’s internal representations—specifically Personalized Initial Knowledge ($\mathbf{l}_c$) and Personalized Learning Rate ($\mathbf{t}_c$) as defined in Section 3.3—faithfully represent the educational constructs postulated by the reference model, we evaluate the representational alignment through two primary metrics that quantify the preservation and functional utility of pedagogical constructs.

1. *Convergent Validity (Representational Alignment)*: This metric, denoted as $\mathcal{I}_1$, measures the statistical correspondence between gTransformer’s latent projections and the reference model’s theoretical parameters. We calculate the Pearson correlation between the student-level estimations and the BKT-derived priors for both initial mastery and learning rate. The metrics are expressed as:

$$\mathcal{I}_{1l} = \text{Corr}(p_{\text{know}}, P(L_0)) \quad \text{and} \quad \mathcal{I}_{1t} = \text{Corr}(p_{\text{rate}}, P(T))$$

(11)

- where $p_{know} = \sigma(\bar{l}_c)$ and $p_{rate} = \sigma(\bar{t}_c)$ are the mean values across the latent embedding dimensions projected through a sigmoid activation.
2. *Predictor Equivalence (Functional Substitutability)*: This metric, denoted as $\mathcal{I}_2$, assesses whether the learned gTransformer parameters can functionally substitute the reference parameters in a simulation. We "inject" the projected factors (p_{know} and p_{rate}) into the canonical BKT recurrence equations to generate *induced mastery trajectories*. $\mathcal{I}_2$ is defined as the correlation between these induced trajectories ($P(L)_{ind}$) and the reference trajectories ($P(L)_{ref}$) generated by the original BKT model:

$$\mathcal{I}_2 = \text{Corr}\left(P(L)_{ind}, P(L)_{ref}\right)$$

(12)

Table 2 summarizes the alignment metrics for the unconstrained gTransformer model ($\lambda_{\mathcal{I}} = 0$), demonstrating that without explicit grounding, the deep model’s latent space remains decoupled from pedagogical constructs.

Table 2. Interpretability Alignment Metrics for unconstrained gTransformer ($\lambda_{\mathcal{I}} = 0$).

Dataset	$\mathcal{I}_{1l}$	Int.	$\mathcal{I}_{1t}$	Int.	$\mathcal{I}_2$	Int.
AS2009	-0.1382	Poor	-0.0067	Negl.	0.1870	Poor
AS2015	-0.0392	Negl.	0.0908	Negl.	0.0975	Negl.
Bridge2006	-0.0645	Negl.	0.0160	Negl.	0.0809	Negl.
NIPS34	0.2070	Poor	-0.0425	Negl.	0.1722	Poor

Enabling Representational Grounding ($\lambda_{\mathcal{I}} = 0.10$) leads to a significant restoration of semantic alignment, as evidenced in Table 3.

Table 3. Interpretability Alignment Metrics for grounded gTransformer ($\lambda_{\mathcal{I}} = 0.1$).

Dataset	$\mathcal{I}_{1l}$	Int.	$\mathcal{I}_{1t}$	Int.	$\mathcal{I}_2$	Int.
AS2009	0.5409	Good	0.3131	Fair	0.6250	Excell.
AS2015	0.9217	Excell.	0.8801	Excell.	0.1749	Poor
Bridge2006	0.4561	Fair	0.6422	Good	0.0859	Negl.
NIPS34	0.4912	Fair	0.4105	Fair	0.2104	Poor

The results presented in Tables 2 and 3 reveal a notable divergence between *Convergent Validity* ($\mathcal{I}_1$) and *Predictor Equivalence* ($\mathcal{I}_2$). Although $\mathcal{I}_1$ attains high levels across most datasets even with relatively low $\lambda_{\mathcal{I}}$ grounding weights, $\mathcal{I}_2$ remains consistently low.

High $\mathcal{I}_1$ scores demonstrate that gTransformer successfully internalizes the *semantic identity* of the theoretical constructs (Initial Knowledge and Learning Rate). However, $\mathcal{I}_2$ measures functional substitutability within a comparatively constrained reference model that is unable to capture the intricate behavioral patterns identified by gTransformer. As a high-capacity Transformer, gTransformer captures complex, long-range dependencies and context-aware dynamics that exceed the modeling capability of classical BKT. This discrepancy provides empirical evidence that gTransformer does not merely mimic the reference model, but rather maps its core pedagogical constructs onto a more sophisticated and predictive architecture.

5.2.1. Latent Structure Validation via Diagnostic Probing

While the convergent validity metrics ($\mathcal{I}_1, \mathcal{I}_2$) provide evidence that the model’s inputs are successfully grounded, a question remains: does this pedagogical signal survive the Transformer’s deep processing? To verify this, we employ *Diagnostic Probing with Control Tasks* [40] to validate the *Structural Encoding* of constructs at the point of prediction.

The probing protocol investigates whether the theoretical constructs injected into the embedding layer (Equations ?? and ??) remain dominant within the final contextualized

representations (z_t) used for performance prediction. If the information remains linearly recoverable from z_t , it proves that the Transformer built its predictive logic upon the grounded foundations.

To validate this, we extract the task-contextualized representations (z_t)—the concatenation of the decoder’s hidden state and the task embedding—and train a simple linear regression probe to recover the BKT mastery probability $P(L_t)$. However, high performance on this *true task* alone is insufficient, as high-capacity models might memorize arbitrary patterns. To rigorously distinguish genuine encoding from superficial correlation, we implement a *control task* [40] by training an identical probe to predict a randomly shuffled version of the theoretical labels. We quantify the integrity of the representation via the *Selectivity* metric:

$$\text{Selectivity} = R^2_{\text{true}} - R^2_{\text{control}}$$

(13)

where R^2_{true} measures the variance explained when predicting actual BKT mastery, and R^2_{control} measures the same for the shuffled control. A high positive selectivity score ($\Delta R^2 > 0.5$) provides terminal evidence of *Structural Encoding*, confirming that the pedagogical constructs remain the primary informative features through the model’s entire reasoning chain.

Table 4 presents the probing results for the grounded gTransformer model ($\lambda_{\mathcal{I}} = 0.1$) on two representative datasets.

Table 4. Diagnostic Probing Results for grounded gTransformer ($\lambda_{\mathcal{I}} = 0.1$).

Dataset	R^2_{true}	R^2_{control}	Selectivity (ΔR^2)	Interpretation
AS2009	0.6731	-0.0139	0.6870	Robust Structural Encoding
AS2015	0.6472	-0.0043	0.6515	Robust Structural Encoding

The results provide compelling validation of gTransformer’s interpretability. Both datasets exhibit selectivity scores exceeding 0.65, substantially above the threshold for “strong encoding” ($\Delta R^2 > 0.5$) established in the literature [40]. The slightly negative control task performance confirms that the model is not simply memorizing arbitrary patterns, but has genuinely organized its post-attention representations around meaningful constructs. Critically, these probing results validate that the grounded information embedded in the input layer (via l_c and t_c) is preserved through the Transformer’s self-attention mechanism and remains structurally encoded in the final contextualized representations. This demonstrates that the representational grounding fundamentally shapes how the model processes and represents student knowledge states.

This dual validation—combining correlation-based semantic alignment ($\mathcal{I}_1, \mathcal{I}_2$) with rigorous probing selectivity—establishes that gTransformer satisfies the requirements of Pedagogical Interpretability defined in Section 2.3. The model’s representations are both semantically meaningful (aligned with BKT constructs) and structurally validated via control tasks, providing a robust foundation for pedagogically interpretable predictions.

5.3. Accuracy–Interpretability Trade-Off (RQ2)

A common assumption in deep knowledge tracing is that predictive performance and interpretability are opposing goals. To explore this relationship, we construct a *Pareto frontier* by systematically varying the *Alignment Intensity* ($\lambda_{\mathcal{I}}$), a unified hyperparameter that scales the weights of all three interpretability loss components simultaneously: $\lambda_{ref} = \lambda_{know} = \lambda_{rate} = \lambda_{\mathcal{I}}$. By varying $\lambda_{\mathcal{I}} \in [0, 1]$, we map the resulting effects on both accuracy and alignment.

We measure the overall interpretability of the model using a *Composite Alignment* metric ($\bar{\mathcal{I}}$), defined as the arithmetic mean of the representational ($\mathcal{I}_{1l}$, $\mathcal{I}_{1t}$) and functional ($\mathcal{I}_2$) components:

$$\bar{\mathcal{I}} = \frac{1}{3}(\mathcal{I}_{1l} + \mathcal{I}_{1t} + \mathcal{I}_2)$$

(14)

Tables 5 and 6 summarize the results for the AS2009 and AS2015 datasets across the $\lambda_{\mathcal{I}}$ sweep.

Figures 3 and 4 illustrate the resulting frontiers. We observe that Representational Grounding serves as an effective regularizer, although the *Saturation Point*—the weight at which interpretability is maximized without significant loss of fidelity—depends on dataset complexity. For the problem-level AS2009 dataset, we observe a synergy point at $\lambda_{\mathcal{I}} = 0.10$ where both metrics improve. Conversely, the concept-level AS2015 dataset reaches saturation at lower constraints, retaining 99.9% of baseline AUC. Beyond these points, a *grounding collapse* starts, where excessive theoretical constraints force the model to ignore nuanced behavioral signals in favor of rigid theoretical priors.

Table 5. Pareto Sweep results for AS2015.

$\lambda_{\mathcal{I}}$	AUC	Alignment ($\bar{\mathcal{I}}$)	Interpretation
0.0 (Baseline)	0.7248	-0.0012	Black Box
0.1 (Saturation)	0.7245	0.6673	99.9% AUC Retained
0.2	0.7181	0.6453	High-Fidelity Diagnostic
0.3	0.7112	0.6095	Latent Degradation
0.5	0.6975	0.5691	Theory-Dominant
0.8	0.6830	0.4828	Grounding Collapse
1.0	0.6763	0.4828	Grounding Collapse

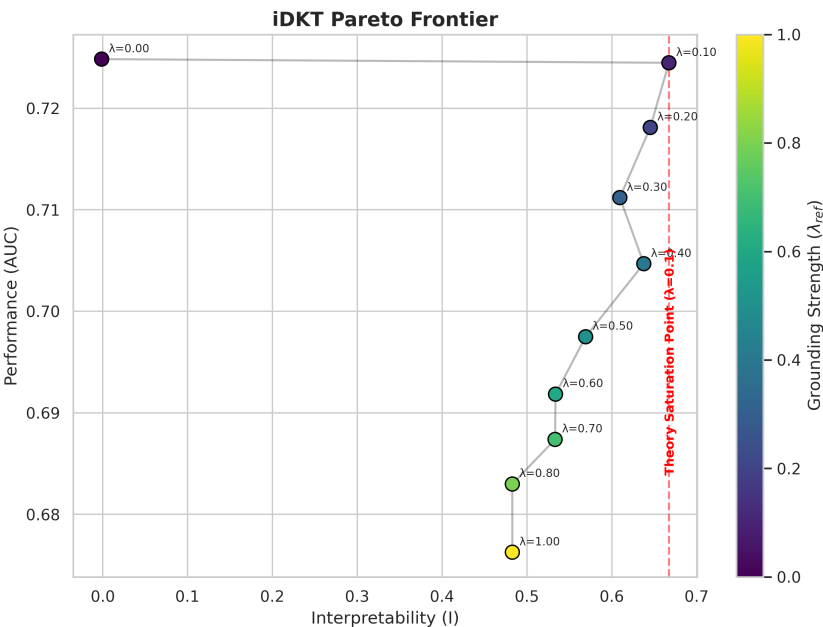


Figure 3. gTransformer Pareto Frontier for AS2015. The trajectory illustrates a saturation of theoretical alignment at $\lambda_{\mathcal{I}} \approx 0.1$, reaching maximum interpretability before performance degrades.

5.4. Context-Aware Diagnostics (RQ2)

In the following sections, we analyze how the architectural design of gTransformer can be leveraged to provide context-aware diagnostics, addressing the "Diagnostic Gap" identified in RQ2.

Table 6. Pareto Sweep results for AS2009.

$\lambda_{\mathcal{I}}$	AUC	Alignment ($\tilde{\mathcal{I}}$)	Interpretation
0.0 (Baseline)	0.8207	0.0293	Black Box
0.1 (Synergy)	0.8255	0.3846	Accuracy-Interpretability Synergy
0.2	0.8110	0.4270	Theory-Balanced Bias
0.3 (Saturation)	0.7963	0.4790	Theoretical Saturation
0.5	0.7760	0.4125	Grounding Degradation
0.8	0.7552	0.0708	Grounding Collapse
1.0	0.7501	0.0714	Grounding Collapse

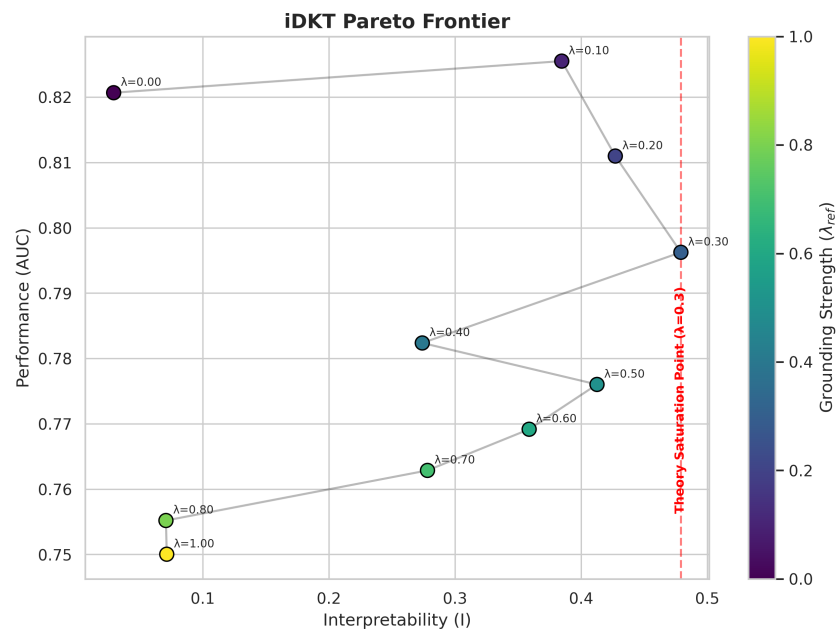


Figure 4. gTransformer Pareto Frontier for AS2009. The curve displays an accuracy peak at $\lambda_{\mathcal{I}} = 0.1$, followed by a non-monotonic transition toward saturation and eventual collapse as constraints overwhelm behavioral patterns.

5.4.1. Contextualization

The capability of gTransformer to leverage longitudinal context is illustrated in Figure 5, which displays a "Twin Divergence" for a specific skill (Skill 4: Area Circle) in the AS2009 dataset. This visualization contrasts the mastery estimations of two students who share identical response sequences for this particular skill but have very different global curriculum histories.

While a Markovian model like BKT is mathematically forced to assign identical mastery states to both students due to their identical local sequences, gTransformer leverages its self-attention mechanism to provide high-resolution contextual diagnostics through two distinct phenomena:

- *Non-Markovian Initialization:* At the very first interaction ($t = 0$), gTransformer already provides divergent estimations. Student 2208 is recognized as having a significantly higher foundational baseline for this concept than Student 167, despite both having the same immediate local history.
- *Velocity-Driven Cross-Over:* Despite starting from a lower baseline, Student 167 exhibits a steeper learning curve (higher v_s). After the first success, their mastery estimation increases rapidly, eventually overtaking Student 2208 at the "Cross-Over Point" (approximately $t = 1.5$).

This dual evidence demonstrates that gTransformer effectively decouples *Initial Standing* from *Learning Velocity*. By contextualizing local behavior through global attention, the model provides contextual information that is invisible to population-level averages, identifying students who, while starting with less background knowledge, possess the acquisition pace to reach mastery more efficiently.

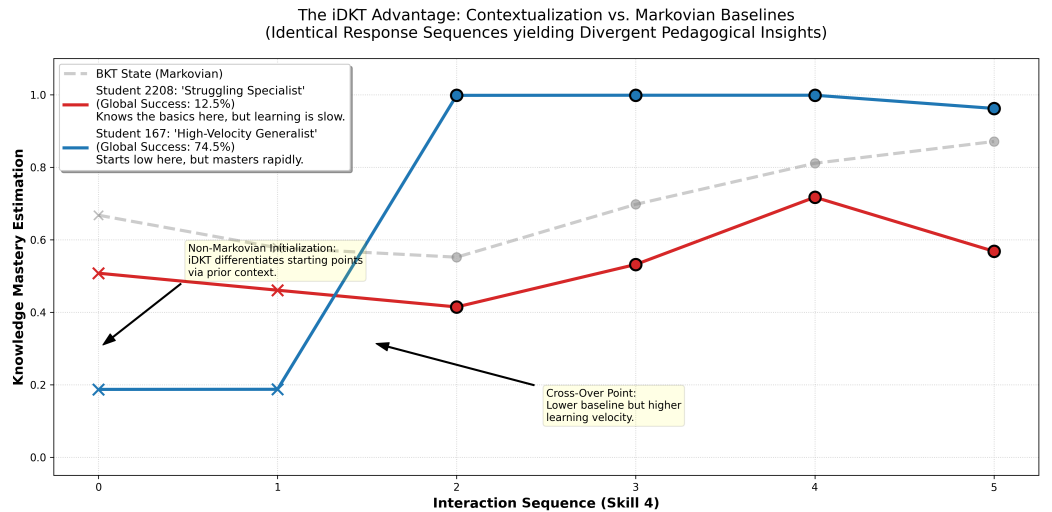


Figure 5. Twin Sequence Divergence for Skill 4 in AS2009. Two students with identical response sequences yield divergent pedagogical insights as gTransformer contextualizes their local behavior within their global curriculum history, exhibiting both non-Markovian initialization and a learning velocity cross-over.

5.4.2. Individualization and Behavioral Heterogeneity

The primary contribution of gTransformer, and the core of RQ2, lies in its ability to transform population-level theoretical averages into high-granularity contextual diagnostics. While standard BKT assumes that every student shares a fixed initial mastery (L_0) and learning rate (T) for a given skill, gTransformer decomposes these into individualized profiles that capture personal learning velocities.

As illustrated in Figure 6, gTransformer maps the student population into a continuous *diagnostic manifold* defined by two student-specific parameters: the Knowledge Gap (k_c , representing placement) and the Learning Velocity (v_s , representing pacing). Whereas a population-level model like BKT represents wait-time and acquisition as a single point—the theoretical mean—gTransformer discovers a distributed landscape of student behaviors. Through unsupervised clustering, we identify four distinct learning archetypes: **Initial Standing** (Low k_c , Low v_s), **Developing Pacing** (Low k_c , Moderate v_s), **High-Velocity Growth** (Low k_c , High v_s), and **Advanced Standing** (High k_c , High v_s).

To statistically quantify this granularity, we analyze the *delta distribution* ($\Delta = p_{rate} - P(T)$), representing the student-specific deviation from the theoretical learning rate across all skills. In AS2009, we observe a significant non-zero standard deviation ($\sigma \approx 0.019$). This represents a fundamental increase in diagnostic resolution compared to the baseline BKT, where $\sigma = 0$ by definition. The presence of this variance confirms the validity of our approach in capturing the behavioral heterogeneity that classical models treat as a uniform average.

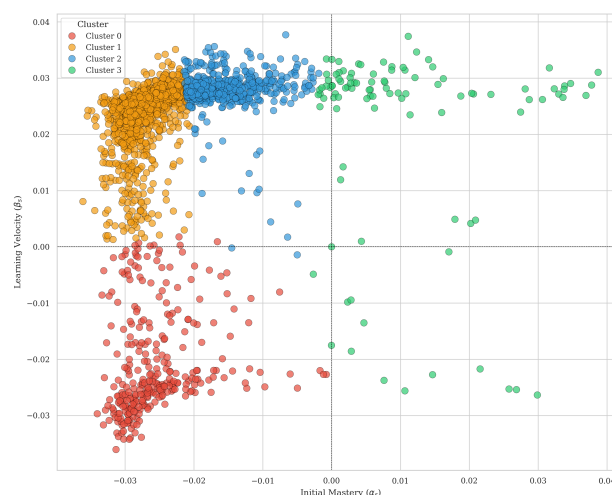


Figure 6. Discovery of student archetypes in the gTransformer diagnostic space. Students are mapped along individualized Knowledge Gap (k_c) and Learning Velocity (v_s) axes, revealing the behavioral heterogeneity hidden within population-level BKT averages.

5.4.3. High-Granularity Trajectory Analysis

To provide visual evidence for RQ2, we generated the mosaic showed in Figure 7 displaying individualized trajectories for three representative individual students (Fast, Median, Slow) per skill across the 15 most frequent skills in the dataset (including major topics like Circle Graphs, Venn Diagrams, and Linear Equations).

The qualitative analysis of these trajectories reveals three significant pedagogical insights:

1. **Dynamic Diagnostic Placement:** Even when BKT is forced to start at the population prior (L_0), gTransformer starting points ($t = 1$) vary per student. This demonstrates that gTransformer leverages cross-skill behavioral transfers to perform individualized placement before the first interaction with a new skill.
2. **Heterogeneous Learning Velocities:** We observe distinct slope gradients across students, confirming gTransformer's ability to model individualized acquisition rates (t_s). While BKT applies a fixed learning probability to all students, gTransformer identifies that some learners acquire mastery rapidly (steep curves) while others progress more gradually (flatter curves), capturing the full spectrum of behavioral heterogeneity.

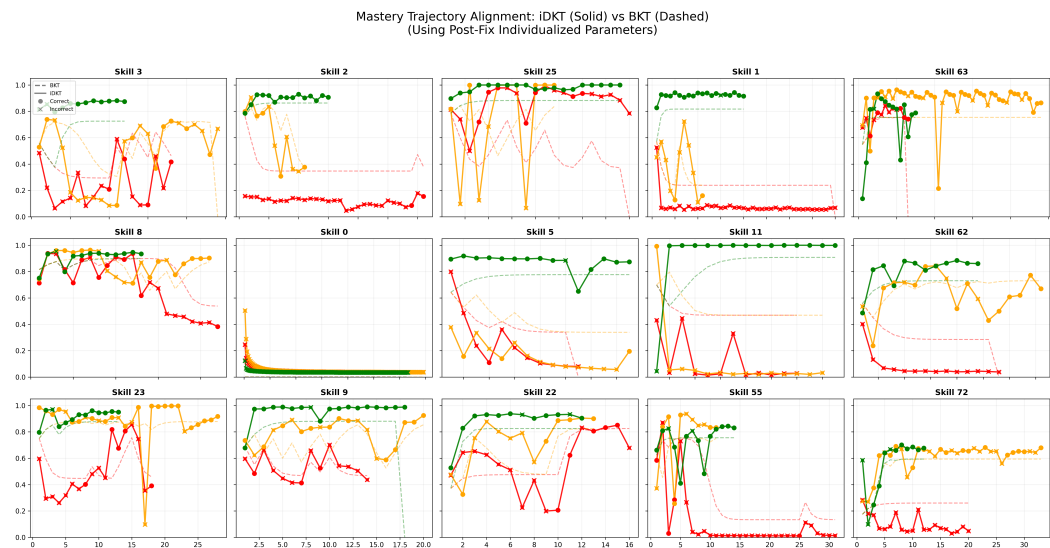


Figure 7. Mastery Trajectory Alignment Mosaic for AS2009. Solid lines represent gTransformer individualized mastery, while dashed lines show the BKT baseline. Markers indicate observed student performance (Circle: Correct, X: Incorrect). Each plot displays three unique students selected as archetypal representatives of Fast, Median, and Slow learning behaviors (non-aggregated).

5.4.4. Archetype-Level Diagnostics

Individualized student profiling offers new opportunities for pedagogical monitoring and adaptive intervention. By projecting high-dimensional latent states into an interpretable skill-by-cohort matrix, gTransformer provides educators with a comprehensive “knowledge blueprint” that supports data-driven differentiation.

This utility is visualized in Figure 8, which displays the mean latent mastery predicted by gTransformer for four student clusters across 30 curriculum skills. The map reveals distinct “knowledge ceilings” and “learning frontiers” for each group, proving that the granularity discovered by the model translates into actionable insights for targeted instructional design. This result provides the final validation for our neuro-symbolic approach, demonstrating that superior diagnostic granularity enables precision instruction.

5.4.5. Pedagogical Confidence

A significant practical application of this interpretability is evaluating the reliability of model predictions. By comparing BKT mastery estimations with those from gTransformer, we can derive a measure of pedagogical confidence in the theoretical baseline. This relationship is visualized in Figure 9, which displays the concordance between BKT and gTransformer mastery predictions across the most frequent skills and students.

The AS2015 results demonstrate high levels of agreement, whereas the AS2009 data reveals more divergence, where gTransformer uncovers individualized patterns that deviate from the theoretical prior. These divergences serve as critical diagnostic indicators, identifying students or curriculum segments where the population-level model may be insufficient and personalized intervention is advisable. This result provides further validation for RQ2, demonstrating that superior diagnostic granularity directly translates into actionable pedagogical insights.

6. Conclusions

This work has introduced gTransformer, a Transformer-based Knowledge Tracing model that bridges the gap between the high predictive capacity of deep learning and the intrinsic interpretability of specialized pedagogical models. By utilizing Representational

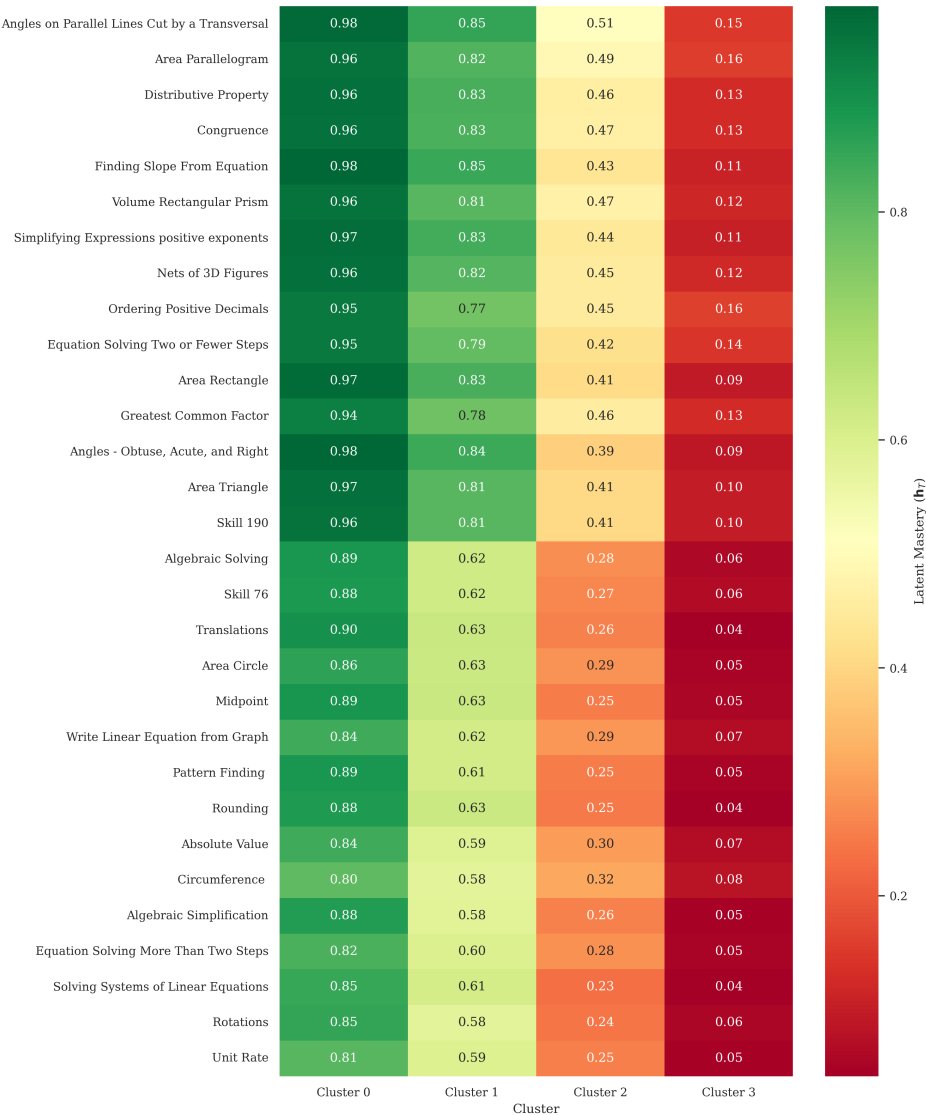
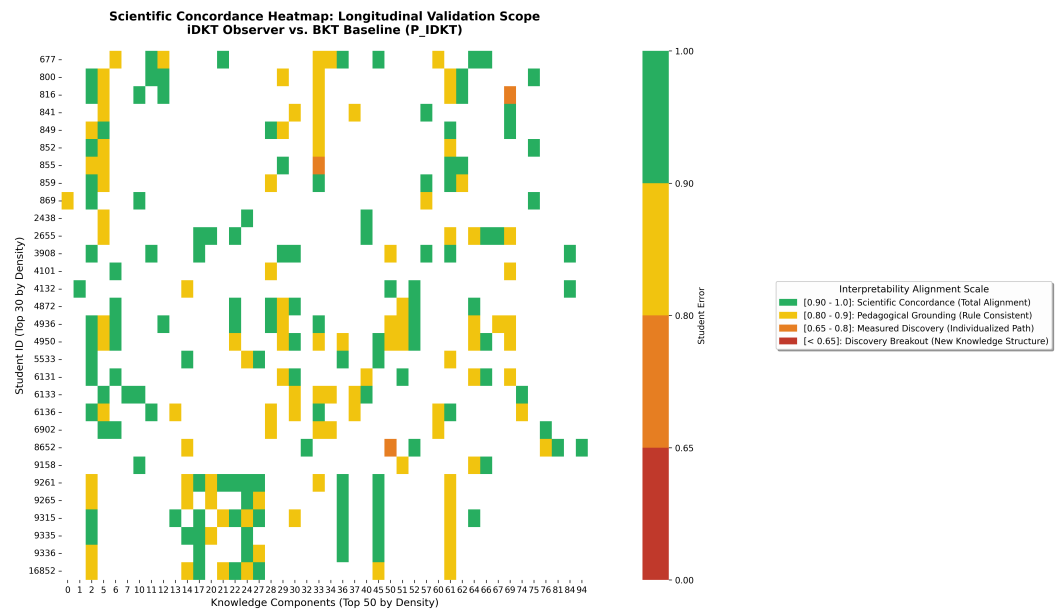
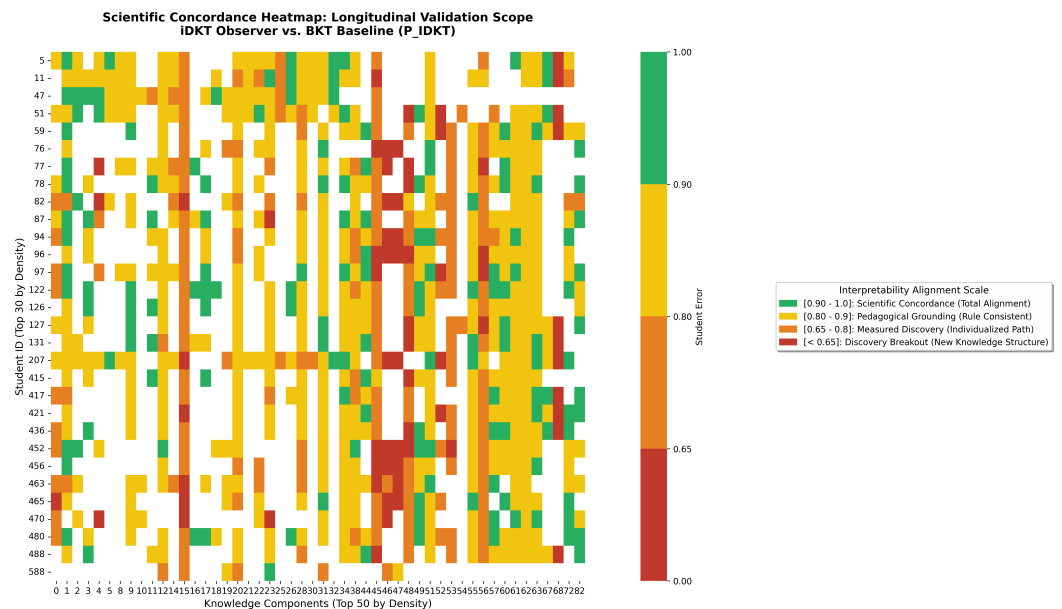


Figure 8. Pedagogical Diagnostic Map for AS2009. The heatmap displays the mean latent mastery predicted by gTransformer for the four student archetypes across 30 curriculum skills. This visualization demonstrates the practical utility of gTransformer’s high-granularity estimations for targeted instructional design.



(a) AS2015



(b) AS2009

Figure 9. Heatmaps of per-skill prediction alignment.

Grounding, we have demonstrated that it is possible to anchor deep latent representations to semantically meaningful constructs without sacrificing state-of-the-art predictive performance.

Our experimental results successfully address the proposed research questions. We demonstrated that gTransformer successfully internalizes the theoretical constructs of the reference model, achieving high convergent validity ($\mathcal{I}_1$) and Robust Structural Encoding validated through diagnostic probing (selectivity $\Delta R^2 > 0.65$). This dual validation establishes that the semantic alignment reflects a genuine internal representation of knowledge states rather than superficial pattern matching. Furthermore, we formalized a methodology for exploring the accuracy–interpretability trade-off, identifying synergistic points where significant alignment is achieved with minimal impact on predictive AUC. Finally, we proved that gTransformer provides a substantial increase in diagnostic granularity compared to population-level baselines, identifying individualized learning velocities that enable truly adaptive educational pacing.

The contribution of this research is twofold: a formal validation framework for evaluating the internal interpretability of deep learning models in education, and a practical architecture that transforms complex predictors into actionable pedagogical tools. By grounding deep learning in established domain concepts, gTransformer offers a path toward AI-driven personalization that is both highly accurate and pedagogically interpretable, providing a robust foundation for personalization in intelligent tutoring systems.

Author Contributions: Conceptualization, C. L.; methodology, C. L. and O. C. S.; software, C. L.; validation, C. L. and O. C. S.; formal analysis, C. L.; investigation, C. L.; resources, C. L. and O. C. S.; data curation, C. L.; writing—original draft preparation, C. L.; writing—review and editing, C. L. and O. C. S.; visualization, C. L.; supervision, O. C. S.; project administration, O. C. S.; funding acquisition, O. C. S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The datasets used in this paper can be downloaded through the links provided in <https://pykt-toolkit.readthedocs.io/en/latest/datasets.html>.

Conflicts of Interest: The authors declare no conflicts of interest.

References

- Corbett, A.T.; Anderson, J.R. Knowledge tracing: Modeling the acquisition of procedural knowledge. *User modeling and user-adapted interaction* **1994**, *4*, 253–278.
- Abdelrahman, G.; Wang, Q.; Nunes, B. Knowledge tracing: A survey. *ACM Computing Surveys* **2023**, *55*, 1–37.
- Cen, H.; Koedinger, K.; Junker, B. Comparing two IRT models for conjunctive skills. In *Proceedings of the International conference on intelligent tutoring systems*. Springer, 2008, pp. 796–798.
- Pavlik Jr, P.I.; Cen, H.; Koedinger, K.R. Performance Factors Analysis—A New Alternative to Knowledge Tracing. *Online submission* **2009**.
- Vie, J.J.; Kashima, H. Knowledge tracing machines: Factorization machines for knowledge tracing. In *Proceedings of the Proceedings of the AAAI conference on artificial intelligence*, 2019, Vol. 33, pp. 750–757.
- Embretson, S.E.; Reise, S.P. *Item response theory for psychologists*; Psychology Press, 2013.
- Šarić Grgić, I.; Grubišić, A.; Gašpar, A. Twenty-five years of Bayesian knowledge tracing: a systematic review, 2022.
- Piech, C.; Bassen, J.; Huang, J.; Ganguli, S.; Sahami, M.; Guibas, L.J.; Sohl-Dickstein, J. Deep knowledge tracing. *Advances in neural information processing systems* **2015**, *28*.
- Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention is all you need. *Advances in neural information processing systems* **2017**, *30*.
- Bai, Y.; Zhao, J.; Wei, T.; Cai, Q.; He, L. A survey of explainable knowledge tracing. *Applied Intelligence* **2024**, *54*, 6483–6514.
- Pandey, S.; Karypis, G. A self-attentive model for knowledge tracing. *International Educational Data Mining Society*, 2019, p. 384–389. Cited by: 270.

12. Ghosh, A.; Heffernan, N.; Lan, A.S. Context-aware attentive knowledge tracing. In Proceedings of the Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining, 2020, pp. 2330–2339. 701
13. Yeung, C.K. Deep-IRT: Make deep learning based knowledge tracing explainable using item response theory. *arXiv preprint arXiv:1904.11738* **2019**. 702
14. Su, Y.; Cheng, Z.; Luo, P.; Wu, J.; Zhang, L.; Liu, Q.; Wang, S. Time-and-concept enhanced deep multidimensional item response theory for interpretable knowledge tracing. *Knowledge-Based Systems* **2021**, *218*, 106819. 703
15. Chen, J.; Liu, Z.; Huang, S.; Liu, Q.; Luo, W. Improving interpretability of deep sequential knowledge tracing models with question-centric cognitive representations. In Proceedings of the Proceedings of the AAAI conference on artificial intelligence, 2023, Vol. 37, pp. 14196–14204. 704
16. Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence* **2019**, *1*, 206–215. 705
17. Von Rueden, L.; Mayer, S.; Beckh, K.; Georgiev, B.; Giesselbach, S.; Heese, R.; Kirsch, B.; Pfrommer, J.; Pick, A.; Ramamurthy, R.; et al. Informed Machine Learning – A Taxonomy and Survey of Integrating Knowledge into Learning Systems. *IEEE Transactions on Knowledge and Data Engineering* **2021**, p. 1–1. 706
18. Yudelson, M.V.; Koedinger, K.R.; Gordon, G.J. Individualized bayesian knowledge tracing models. In Proceedings of the International conference on artificial intelligence in education. Springer, 2013, pp. 171–180. 707
19. Sutskever, I. Training recurrent neural networks. PhD thesis, 2013. 708
20. Choi, Y.; Lee, Y.; Cho, J.; Baek, J.; Kim, B.; Cha, Y.; Shin, D.; Bae, C.; Heo, J. Towards an appropriate query, key, and value computation for knowledge tracing. In Proceedings of the Proceedings of the seventh ACM conference on learning@ scale, 2020, pp. 341–344. 709
21. Bach, S.; Binder, A.; Montavon, G.; Klauschen, F.; Müller, K.R.; Samek, W. On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PloS one* **2015**, *10*, e0130140. 710
22. Ribeiro, M.T.; Singh, S.; Guestrin, C. "Why should i trust you?" Explaining the predictions of any classifier. In Proceedings of the Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, 2016, pp. 1135–1144. 711
23. Gan, W.; Sun, Y.; Sun, Y. Knowledge interaction enhanced knowledge tracing for learner performance prediction. In Proceedings of the 2020 7th International conference on behavioural and social computing (BESC). IEEE, 2020, pp. 1–6. 712
24. Zhou, Y.; Li, X.; Cao, Y.; Zhao, X.; Ye, Q.; Lv, J. LANA: towards personalized deep knowledge tracing through distinguishable interactive sequences. *arXiv preprint arXiv:2105.06266* **2021**. 713
25. Liu, S.; Yu, J.; Li, Q.; Liang, R.; Zhang, Y.; Shen, X.; Sun, J. Ability boosted knowledge tracing. *Information Sciences* **2022**, *596*, 567–587. 714
26. Sun, J.; Wei, M.; Feng, J.; Yu, F.; Li, Q.; Zou, R. Progressive knowledge tracing: Modeling learning process from abstract to concrete. *Expert Systems with Applications* **2024**, *238*, 122280. 715
27. Zhang, L. Learning factors knowledge tracing model based on dynamic cognitive diagnosis. *Mathematical Problems in Engineering* **2021**, *2021*, 8777160. 716
28. Zhu, J.; Yu, W.; Zheng, Z.; Huang, C.; Tang, Y.; Fung, G.P.C. Learning from interpretable analysis: Attention-based knowledge tracing. In Proceedings of the International conference on artificial intelligence in education. Springer, 2020, pp. 364–368. 717
29. Lee, U.; Park, Y.; Kim, Y.; Choi, S.; Kim, H. Monacobert: Monotonic attention based convbert for knowledge tracing. In Proceedings of the International Conference on Intelligent Tutoring Systems. Springer, 2024, pp. 107–123. 718
30. Zhang, M.; Zhu, X.; Zhang, C.; Qian, W.; Pan, F.; Zhao, H. Counterfactual Monotonic Knowledge Tracing for Assessing Students' Dynamic Mastery of Knowledge Concepts. In Proceedings of the Proceedings of the 32nd ACM international conference on information and knowledge management, 2023, pp. 3236–3246. 719
31. Karpatne, A.; Atluri, G.; Faghmous, J.; Steinbach, M.; Banerjee, A.; Ganguly, A.; Shekhar, S.; Samatova, N.; Kumar, V. Theory-guided Data Science: A New Paradigm for Scientific Discovery from Data. *IEEE Transactions on Knowledge and Data Engineering* **2017**, *29*, 2318–2331. 720
32. Karniadakis, G.E.; Kevrekidis, I.G.; Lu, L.; Perdikaris, P.; Wang, S.; Yang, L. Physics-informed machine learning. *Nature Reviews Physics* **2021**, *3*, 422–440. 721
33. Jia, X.; Willard, J.; Karpatne, A.; Read, J.S.; Zwart, J.A.; Steinbach, M.; Kumar, V. Physics-guided machine learning for scientific discovery: An application in simulating lake temperature profiles. *ACM/IMS Transactions on Data Science* **2021**, *2*, 1–26. 722
34. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics* **2019**, *378*, 686–707. 723
35. Willard, J.; Jia, X.; Xu, S.; Steinbach, M.; Kumar, V. Integrating scientific knowledge with machine learning for engineering and environmental systems. *ACM Computing Surveys* **2022**, *55*, 1–37. 724
36. Liu, Z.; Liu, Q.; Chen, J.; Huang, S.; Tang, J.; Luo, W. pyKT: a python library to benchmark deep learning based knowledge tracing models. *Advances in Neural Information Processing Systems* **2022**, *35*, 18542–18555. 725

37. Feng, M.; Heffernan, N.; Koedinger, K. Addressing the assessment challenge with an online system that tutors as it assesses. *User modeling and user-adapted interaction* **2009**, *19*, 243–266. 755
 38. Stamper, J.; Niculescu-Mizil, A.; Ritter, S.; Gordon, G.; Koedinger, K. Bridge to algebra 2006-2007. *Development data sets from KDD Cup* **2010**. 756
 39. Wang, Z.; Lamb, A.; Saveliev, E.; Cameron, P.; Zaykov, Y.; Hernández-Lobato, J.M.; Turner, R.E.; Baraniuk, R.G.; Barton, C.; Jones, S.P.; et al. Instructions and guide for diagnostic questions: The neurips 2020 education challenge. *arXiv preprint arXiv:2007.12061* **2020**. 757
 40. Hewitt, J.; Liang, P. Designing and Interpreting Probes with Control Tasks. In Proceedings of the Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP); Inui, K.; Jiang, J.; Ng, V.; Wan, X., Eds., Hong Kong, China, 2019; pp. 2733–2743. 758
- <https://doi.org/10.18653/v1/D19-1275>. 760

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content. 761